



An efficient cold start strategy for proton exchange membrane fuel cell stacks

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HIGHLIGHTS

- The metal bipolar plate PEMFC stack realized self-starting from $-20\text{ }^{\circ}\text{C}$ within 30 s.
- This strategy combines the advantages of self-cold start and assisted cold start.
- This strategy can achieve a rapid, safe and efficient cold start down to $-40\text{ }^{\circ}\text{C}$.
- Voltage consistency of the stack during cold start was analyzed.

ARTICLE INFO

Keywords:

Cold start strategy
Voltage consistency
Metal bipolar plates
Proton exchange membrane fuel cell stacks

ABSTRACT

Starting the proton exchange membrane fuel cell (PEMFC) stack from subfreezing temperatures is a remaining challenge for the commercialization of PEMFC vehicles. This paper aims to provide an efficient cold start strategy for PEMFC stacks to meet the requirements of rapid, safe and efficient startup in a wide range of subfreezing temperatures. Herein, an efficient cold start strategy combines the advantages of self-starting and coolant heating is proposed and validated in terms of startup time, voltage consistency and energy consumption. Results demonstrate that self-starting from $-20\text{ }^{\circ}\text{C}$ is achieved within 30 s. Rapid and safe cold-start from $-30\text{ }^{\circ}\text{C}$ and $-40\text{ }^{\circ}\text{C}$ is also realized within 82 s and 100 s, respectively. The target heating temperature of $-20\text{ }^{\circ}\text{C}$ is optimal for this strategy. The voltage consistency of the stack during cold start is closely related to cold start performance, and is affected by the starting temperature and the heating power. The stack voltage consistency increases with the increase of starting temperature and heating power. Energy conservation calculations indicate that the heating power should match the stack temperature to avoid excessive energy consumption and insufficient heating. This paper provides an efficient solution for cold starting of a hybrid PEMFC system.

1. Introduction

Achieving carbon neutrality requires vigorous steps to address the increasing carbon dioxide emissions caused by automobile exhaust. It calls for reducing the dependence of transportation on fossil fuels, while providing a green and sustainable development path for the future transportation. In this light, new energy vehicles can provide efficient solutions to this problem and are gaining a considerable share of the automotive market. Among various types of new energy vehicles, battery electric vehicles (BEVs) and proton exchange membrane fuel cell (PEMFC) vehicles (FCVs) are the most concerned owing to their inherent

characteristics of high efficiency, low emission and quick start [1,2]. At present, BEV has achieved large scale commercialization in the automotive industry. However, due to the limited driving range and extended charging time, BEV is unsuitable for commercial vehicles, whereas FCVs could provide a promising technological route to this end [3]. Fuel cell technology has developed rapidly in recent decades, but to make FCVs commercially viable, in addition to further reducing costs, improving service lifetime and accelerating infrastructure construction, measures to improve the adaptability to sub-freezing temperatures are also urgently needed [4]. Under subzero temperatures, a normally working fuel cell will not only avoid the problem of shortening the

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<https://doi.org/10.1016/j.jpowsour.2022.231492>

Received 23 March 2022; Received in revised form 9 April 2022; Accepted 18 April 2022

Available online 29 June 2022

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driving range, but also use the waste heat to heat the cabin. Nevertheless, starting a fuel cell from sub-freezing temperatures is always a challenge as the produced water is easy to freeze and may cause startup failure or damage to the electrode. As a result, the cold start ability has become a critical technical target of fuel cell technology. In this regard, the US Department of Energy (DOE) has set the cold start targets for transportation applications, including a rapid cold start from -20°C to 50% of rated power within 30 s, unassisted start from -30°C and assisted start from -40°C [5].

Fundamental mechanisms of cold start dynamics and ice formation theories have been extensively elucidated [6–15]. The core issue of cold start is the competition between heat production and ice formation. A successful cold start requires the cell to raise the temperature above the freezing point before the cathode catalyst layer is blocked with ice. Tajiri [6] and Tabe [7] demonstrated analytically and experimentally that the cathode catalyst layer is uniformly filled with ice at a lower starting current density while the active ice formation region moves from the membrane side to the gas diffusion layer side at higher current densities. In addition, water is more likely to freeze immediately in the catalyst layer at a very low temperature, while can also diffuse into the gas diffusion layer and flow channel in the state of supercooled water and freeze there at a relatively high temperature [9–11]. Huo [12] took supercooled water into consideration and proposed a more rigorous phase transition system to predict the water transport and phase transition mechanisms during cold start. It was found that majority of product water remains in the cell during cold start [14] and the cathode water storage capacity plays an important role in prolonging the startup duration [16]. In this respect, an effective gas purge is a key step to remove the residual water inside the cell and subsequently increase the chance of a successful startup [17], which is comprehensively reviewed by Amamou [18].

Cold start strategies can be divided into self-cold start and assisted cold start according to the heating sources. Self-cold start completely relies on the heat released by the electrochemical reaction, which is usually achieved by controlling the load. Galvanostatic mode is the most commonly used loading strategy as it is easy to control. However, there is a trade-off between heat generation and ice accumulation. A higher current density is favorable to a rapid startup while a smaller current density sustains a slower ice accumulation [6,19]. Besides, the cell is unable to supply a high current density at the initial stage of the cold start due to low initial water content. In this regard, the potentiostatic mode is superior to the galvanostatic one [20,21] as the potentiostatic mode could better balance the water production and heat generation during the whole startup process, which can not only avoid the problem of excessive current in the initial stage, but also ensure a sufficiently rapid startup. However, constant voltage is unsuitable for practical use, especially for fuel cell stacks. To this end, current ramping strategy proves to be an efficient solution [22]. Self-cold start can also be achieved by controlling the power. Du [23] developed a one-dimensional fuel cell stack model to simulate the maximum power cold start and it was found that the maximum power cold start mode could better balance the heat generation and ice formation, leading to improved cold start survivability than that in the constant voltage and constant current modes. Based on the maximum power mode, Amamou [24] proposed an adaptive cold start strategy, in which the online parameter identification method is integrated into a semi-empirical model to cope with the operating conditions uncertainties and to keep track of the best operating points. The proposed cold start strategy is effective in terms of heating time and energy requirement compared with the galvanostatic and potentiostatic modes. However, self-cold start strategy based on load control is easy to apply and does not require modification of the fuel cell system, but this strategy is only adequate at relatively high temperatures, and irreversible degradation may occur after repeated cold start from a relatively low temperature. Thus, in order to compensate for the lack of startup speed and safety of self-cold start, extra heat can be applied to assist the cold start, which is referred as the assisted cold start.

Extra heat can be provided from various energy sources, such as gas heating [25,26], electric heating [26,27], catalyst H_2/O_2 reaction [28–30] and coolant heating [25,31–33]. Jiao [20,26] numerically investigated the effect of air temperature on cold-start performance and demonstrated that increasing the air temperature has negligible significance in improving the cold start, which is attributed to the low heat capacity and heat transfer coefficient of air. By embedding heating wires into the bipolar plate [27] and end plate [25] and locally heating the cathode and end plate greatly improves the cold start performance. Catalytic hydrogen-oxygen reaction method can release considerable heat and achieve a rapid warm-up even at temperatures below -30°C [28,29]. However, such a method lacks safety and reliability. Coolant heating is more suitable for FCVs, as it is flexible in control and avoids too much modifications to the fuel cell system. Rios [32] did numerous cold start tests from 0°C to -30°C using a 4 kW PEMFC stack with bipolar plates (BP) made of graphite and proposed an efficient thermal management strategy for cold starts. It was found that active thermal management strategy is only adequate for temperature $\geq -15^{\circ}\text{C}$ and a part active thermal management strategy using the heater only until a low temperature, such as -15°C , yields a rapid and safer cold start while improving the energy efficiency. However, the active thermal management is based on potentiostatic mode which is hard to control in such a stack consisting of 40 single cells. Besides, the stack bipolar plates were made of graphite, whose cold start performance are bound to be different from the metal bipolar plates, which is poorly studied in current literatures.

As aforementioned, cold start is a hotspot in PEMFC field and existing experimental researches mainly focus on laboratory-scale fuel cells with bipolar plates made of graphite, and only few studies involve stack-level PEMFC. But to promote the commercialization of PEMFC, cold start researches, at present, should be conducted at the stack level. Besides, even though metal bipolar plates are superior to graphite bipolar plates in terms of fabrication costs, mechanical strength, cold start performance and they are also beneficial to increase the volume power density of the stack [34,35], studies concerning the cold start of metal bipolar plates stack are rare. Moreover, only few studies have paid attention to the consistency of the stack during cold start [36–39], which is also crucial for the cold start and health management of the stack. To this end, this paper aims to examine the cold start of a 7-kW PEMFC stack with metal bipolar plates, and proposes an efficient cold start strategy to meet the requirements of rapid, safe and efficient startup of PEMFC stack in a wide range of subzero temperatures. For that purpose, an integrated cold start strategy combines the advantages of self-cold start and coolant heating assisted cold start is proposed and investigated concerning the effects of the stack temperature, the heating power, the target heating temperature. Furthermore, critical parameters such as temperature, power output, startup time, energy consumption and consistency of the stack during cold start are carefully discussed.

2. Experimental

2.1. Stack and test bench

Cold start experiments were conducted on the test bench depicted in Fig. 1 with a 7-kW PEMFC stack. The stack consists of 46 single cells with an active area of 227 cm^2 , and the bipolar plates are made of stainless steel. The test bench was mainly composed of a gas supply system, a load system, a cooling system, a PEMFC stack and an environment chamber. The cooling system consists of a cold start cooling loop and a standard cooling loop, which are depicted in green and black respectively. The cold start cooling loop consists of a coolant tank, a coolant recirculation pump and an electric heater. It can be expected that although we tried to arrange them as close as possible, the test system is far less integrated than a vehicular fuel cell system. Therefore, an on-board fuel cell system with a comparable level of stack would exhibit better cold start performance. The electric heater was a high-voltage positive temperature

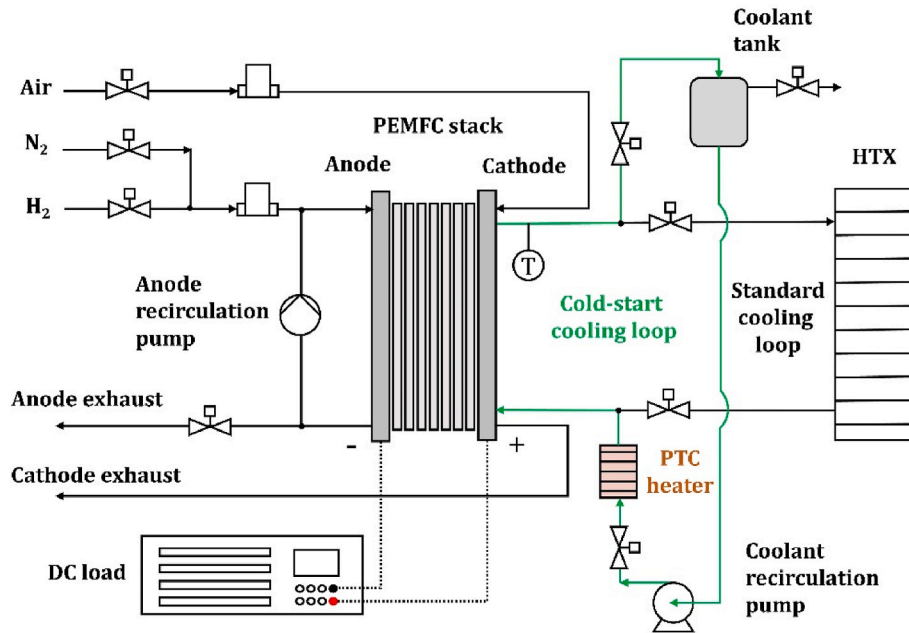


Fig. 1. Schematic of the cold start test bench.

coefficient (PTC) heater with a heating power range of 0–8 kW, which is commonly used in BEVs owing to its characteristics of simple structure, swift response and high safety [40,41]. The load system was equipped with a programmable DC electronic load (Kikusui, PLZ2004) with a maximum load of 10 kW. The stack and cold start cooling loop were located in the environment chamber. The stack temperature was measured at the coolant outlet with a thermocouple (T-type) with the measuring accuracy of $0.5 \pm 0.4\%$ °C. Pure H_2 and air were controlled by mass flowmeters. It should be explained that the gas went through a 2 m long cooling coil in the environment chamber before entering the stack. Thus, considering the heat transfer rate of the gasses, the stack heat capacity and the start-up duration, the influence of the gas temperature is ignored.

2.2. Experimental procedures

Cold start experiments in this study follow the following procedures.

1. Initial conditioning. In order to ensure the stack was identically hydrated before each cold start and to guarantee good consistency and repeatability of the experiment, the stack was operated at a constant discharge current density of 500 mA cm^{-2} for 30 min with 40% humidified air at 60 °C.
2. Purging. The stack was purged at 60 °C with 350 L min^{-1} air and 100 L min^{-1} dry N_2 for 150 s. This step was to remove remaining water inside the stack to avoid the frost damage and guarantee the same initial hydration before freezing.
3. Freezing. The stack and cold-start cooling loop were placed in the environment chamber. The freezing time was 4 h at -10 °C and -15 °C, 6 h at -20 °C and -25 °C, 8 h at -30 °C and -40 °C.
4. Cold start. Cold start was conducted based on current ramping and coolant heating strategies, which will be introduced later. Cold-start cooling loop was in operation and the standard cooling loop was closed during cold start. A successful cold start was defined as the stack output power reaches 50% of its rated power. It should be noted that we use dry air and hydrogen during cold start with no back pressure. For convenience, the flow rate was kept as constant at 330 L min^{-1} and 75 L min^{-1} , respectively, which corresponds to the stoichiometric ratio of 3 and 1.7 under the current density of 600 mA cm^{-2} .

cm^{-2} . Since the loading time is generally within 1 min, a constant gas flow rate is more convenient and acceptable.

2.3. Cold start strategy

The cold start strategy is shown in Fig. 2. This strategy combines both the advantages of self-cold start and coolant heating assisted cold start.

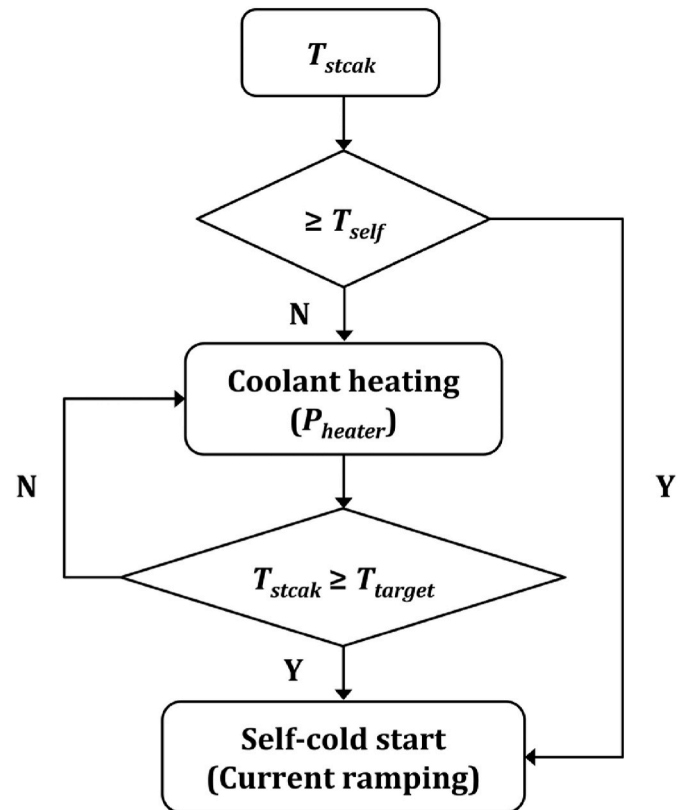


Fig. 2. Block diagram of the cold start strategy (Combines the self-cold start strategy and coolant heating assisted cold start strategy).

First of all, the temperature sensor judges whether the stack temperature (T_{stack}) is greater than the temperature threshold of self-cold start (T_{self}). If so, the stack is self-started according to the set loading rate. Afterwards, if not, the PTC heater will be turned on to heat the antifreeze with the set heating power (P_{heater}) and to heat the stack. Finally, when T_{stack} reaches the target heating temperature (T_{target}), the stack performs the self-cold start according to the set loading rate and the heater is switched off at the same time. Compared with self-starting, the proposed strategy can achieve fast and safe start-up at lower temperatures, thereby relieving the limitation of start-up temperature on self-starting and avoiding freezing damage of the electrode. In addition, compared with traditional coolant heating method, a special small cooling loop is used during cold start. The coolant and the stack are only heated to a temperature that can be self-started instead of approaching or exceeding the freezing point. Therefore, this strategy can guarantee shorter startup time, lower energy consumption, higher safety and provide output power earlier in the initial stage of startup. This strategy is proposed to establish a fast, safe and efficient cold start strategy for PEMFC stacks in a wide range of subfreezing temperatures. For that purpose, the following points should be considered. First, to improve the energy efficiency, the coolant recirculation pump should run at a relatively low speed during the self-cold start process and at a relatively high speed when the PTC heater was operating. Second, current ramping strategy was adopted during self-cold start, which is easier to apply to a PEMFC system than the potentiostatic mode, as it is almost impossible to evenly distribute the voltage to each single cell in the stack in the potentiostatic mode during cold start. Third, T_{self} , P_{heater} and T_{target} need to be determined based on the experimental results, considering the startup time, voltage consistency and energy consumption.

3. Results and discussion

3.1. Self-cold start

Self-start is regarded as the optimal solution for starting a PEMFC stack from subfreezing temperatures. However, self-cold start capability is limited, to some extent, due to the huge heat requirement and the fact that the stack cannot operate with a high current density in a short time under sub-zero temperature conditions [22,42]. This section is going to elucidate the self-cold start performance and characteristics of a 7 kW PEMFC stack and to determine the temperature threshold for self-starting. The evolution of the current density, the stack power and the stack temperature during self-cold start from -10°C , -15°C , -20°C , -25°C and -30°C are shown in Fig. 3. As depicted in Fig. 3(a), current ramping strategy was adopted and the ramping speed for different temperatures were listed in Table 1. It can be seen in Fig. 3(b) that self-cold start from -10°C to -20°C can be achieved within 30 s. Besides, the cold start from -25°C only took 57 s, which directly demonstrates the superior cold start performance of the PEMFC stack with metal bipolar plate. However, even though the temperature difference between -30°C and -25°C is only 5°C , the difficulty of self-starting increases dramatically. The self-cold start from -30°C lasted 233 s until it failed. As shown in Fig. 3(b), the stack power dropped sharply from 213 s. Although the temperature of the coolant outlet had reached 39°C at this time, the startup still failed. The main reason is that the temperature of the coolant drops below zero after passing through the cold start cooling loop, which leads to icing near the inlet, and the icing area gradually spreads downstream. At the same time, the stack cannot sustain such a large current on the limited active area, resulting in the rapid voltage decline, and eventually leads to the failure of the cold start. In addition, as illustrated in Table 1, the stack needs more time to reach the freezing point than the time required to reach 50% of the rated power when self-starting from -10°C to -20°C , while it is equal when starting from -25°C , and the time to reach the freezing point from -30°C is less than the start-up time (even through failed). This can be attributed to the fact that when starting at a higher

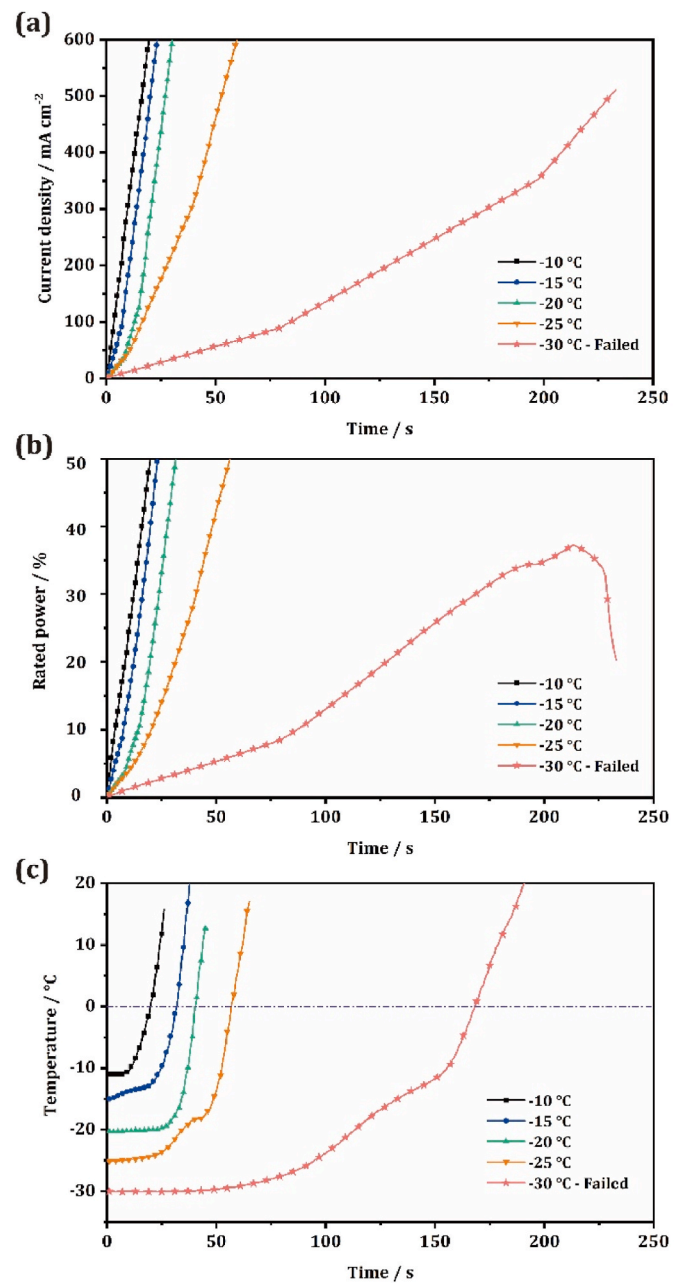


Fig. 3. Self-cold start of PEMFC stack from -10°C , -15°C , -20°C , -25°C and -30°C with current-ramping strategy. (a) evolution of the current density; (b) evolution of the rated power and (c) evolution of the stack temperature.

temperature, the loading rate is large and the coolant flow rate is controlled relatively small, only a small part of the heat is transferred to the coolant. Thereby, most of the heat is absorbed by the membrane electrode assemblies (MEA) and bipolar plates, resulting in less ice formation in the electrode, which indicates that self-cold start is adequate for temperatures greater than -25°C .

To further illustrate the self-cold start characteristics of PEMFC stack with metal BP, voltage distribution and variance were used to evaluate the stack consistency during cold start. As shown in Fig. 4, the voltage distribution was drawn with the minimum voltage, the maximum voltage, the average voltage, the voltage of the NO. 01 cell and the voltage of the NO. 46 cell. The voltage variance was calculated by Eq. (1).

Table 1

Current ramping speed, time to 0 °C and cold start time of self-cold start from −10 °C, −15 °C, −20 °C, −25 °C and −30 °C.

Startup temperature/°C	Current ramping speed/A/s	Time to 0 °C/s	Startup time (Time to 3.5 kW)/s
−10	7 (1 s - end)	20	19
−15	3 (1–8 s)	32	23
	7 (8 s - end)		
−20	1 (1–8 s)	40	30
	3 (8–15 s)		
	7 (15 s - end)		
−25	1 (1–10 s)	57	57
	2 (10–40 s)		
	3 (40 s - end)		
−30	0.25 (1–80 s)	169	Failed
	0.5 (80–200 s)		
	1 (200 s - end)		

$$\delta = \sqrt{\frac{\sum (V_i - V_{avg})^2}{n}} \quad (1)$$

As we know, the lower the startup temperature, the harder the cold start will be. Thus, we cannot apply the same ramping speed for different temperatures. The ramping speed at a lower temperature is obtained by reducing the loading rate in the initial stage on the basis of a higher temperature. Besides, by applying these ramping strategies, the average voltage of the stack during startup shows a consistent trend and similar values (shown in Fig. 4 except for −30 °C case), both of which are between 0.5 and 0.6 V, which justifies our comparison of the voltage consistency. The evolution of the stack average voltage is consistent during self-cold start, showing a trend of rapid decrease first and then slow increase. The main reason is the low initial hydration, the water produced in the initial stage is mainly absorbed by the ionomer in the catalytic layer and membrane [43–45], which facilitates the subsequent increase in voltage. From the point the voltage reaches the minimum value to the end of the start-up, the voltage of each cell shows a monotonous rise, indicating no obvious ice formation in the electrode. In contrast, in the case of failed cold start at −30 °C, it can be clearly seen that the voltages of the NO. 01 cell and the NO. 46 cell show continuous fluctuation, and when the stack is about to fail, the average voltage and the voltage of each single cell drop sharply, indicating that serious freezing have occurred inside the stack [46]. Similar finding was also found by Hu [38], who reported sharp fluctuation of the voltage of the intermediate cells while starting the stack from −25 °C with locally heating the end-plate. Fig. 4 (a)–(e) shows that the uniformity of voltage distribution becomes worse as the startup temperature decrease. It should be noted that the consistency of voltage distribution during normal operation is shown in Fig. S1, and the extreme difference is only 18 mV under the current density of 1300 mA/cm², indicating the stack itself has good consistency. While the exact mechanism for the big difference in voltage consistency between cold start and normal operation remains elusive, possible explanations include: (1) low initial hydration exacerbates the deterioration of stack consistency; (2) the initial water distribution is not uniform and accompanied by a certain randomness, although it has undergone the same purge treatment before cold start, the cell with a slightly higher water content will exhibit a higher voltage; (3) the gases were supplied without back pressure and humidification during cold start; and (4) the cell near the end plates has a larger heat loss, while the cell in the middle of the stack has the smallest heat loss. Luo [39] attributed the worse voltage consistency during cold start to the uneven distribution of reactant gas, generated water and heat in the stack. The fourth reason is end-plate effect [36,47,48], during which the NO. 01 cell and the NO. 46 cell exhibit the lowest voltage and worst cold start performance. In this regard, improving the cold start performance of the cell near the end plate is an effective way to improve the cold start capability of the stack. Nevertheless, for self-starting above −25 °C, using the proposed loading strategy, the minimum voltage during the

start-up is higher than 0.3 V, and the average voltage can also be maintained above 0.5 V, which is quite safe for stack operation [49]. Fig. 4 (f) shows the voltage variance during the self-cold start at different temperatures. The voltage variance first increases rapidly due to the low initial hydration, and the peak value appears later as the temperature decreases, which is ascribed to the fact that the loading rate in the initial stage is slower at lower temperatures. After the voltage variance reaches the maximum value, the voltage variance begins to decrease monotonically for the conditions of −10 °C and −15 °C, indicating that the voltage gradually tends to be uniform during the startup. For −20 °C and −25 °C, the voltage variance decreases for a period of time until a second peak appears, and then decrease monotonously. The main reason is that the startup is fast at −10 °C and −15 °C, and the stack heats up quickly, while at −20 °C and −25 °C, the startup is relatively slow, and the heat exchange between the stack and the coolant is more obvious, causing voltage fluctuations. The final variance at successful startup increases as the temperature decreases, which is obviously reasonable.

3.2. Coolant heating assisted cold start

Coolant heating can be used to assist the cold start of PEMFC, especially for a hybrid FCV. In this section, we discuss the effects of the startup temperature, the heating power and the target heating temperature on the performance of the proposed cold start strategy while taking the factors of startup time, energy consumption and voltage consistency into consideration. The purpose is to provide some practical guidance for the cold start of PEMFC stacks, which can also be used as a supplement to the existing literature.

3.2.1. Cold start from −30 °C

Self-cold start from −30 °C is quite challenging and may cause irreversible decay as illustrated above. However, with the aid of coolant heating, a rapid and safe cold start from −30 °C can be achieved. Fig. 5 shows the cold start process of the stack from −30 °C with the heating power of 4 kW and different target heating temperature of −20 °C, −15 °C and −10 °C. As can be seen in Fig. 5, The PTC heater gradually reaches the set power through PID adjustments. The PTC heater was in operation until the stack reaches the target temperature. Once the PTC heater was turned off, the stack was self-started according to the stack temperature. The coolant outlet temperature did not rise significantly until 35 s after the heater was turned on, and the heating time increases with the target heating temperature under the same heating power as summarized in Table 2, which is apparently reasonable. However, they ended up with almost the same startup time and temperature rise. To explain this, two reasons are considered. First, the heat capacity of the coolant is larger than the stack and it takes more time to heat the coolant itself than the stack. Once the coolant is heated, the temperature of the stack increases significantly. Second, after the heater is switched off, the loading speed for a higher target temperature is larger. The voltage consistency in Fig. 5 (d) shows the same trend as the self-cold start with a first increase and then decrease, the maximum variance appears earlier with a higher T_{target} and the final variance decreases with the increase of the T_{target} , indicating a better voltage consistency. The average voltage during cold start were higher than 0.5 V and the polarization curves (shown in Fig. S3) indicate that the stack performance was well preserved after 10 cold starts from −30 °C, which demonstrates that the proposed strategy is safe for cold start of the stack. However, the differences in voltage consistency and average voltage are too small to be considered, which means increasing the T_{target} doesn't yield significant improvement in cold start performance. Nevertheless, considering the heating time and energy consumption, T_{target} of −20 °C is adequate for cold start from −30 °C.

Fig. 6 shows the effects of heating power (P_{heater}) on the coolant heating assisted cold start of PEMFC stack from −30 °C. The target temperature (T_{target}) was set at −20 °C and the heating power (P_{heater}) were varied from 1.5 kW to 3 kW. Apparently, it can be speculated that

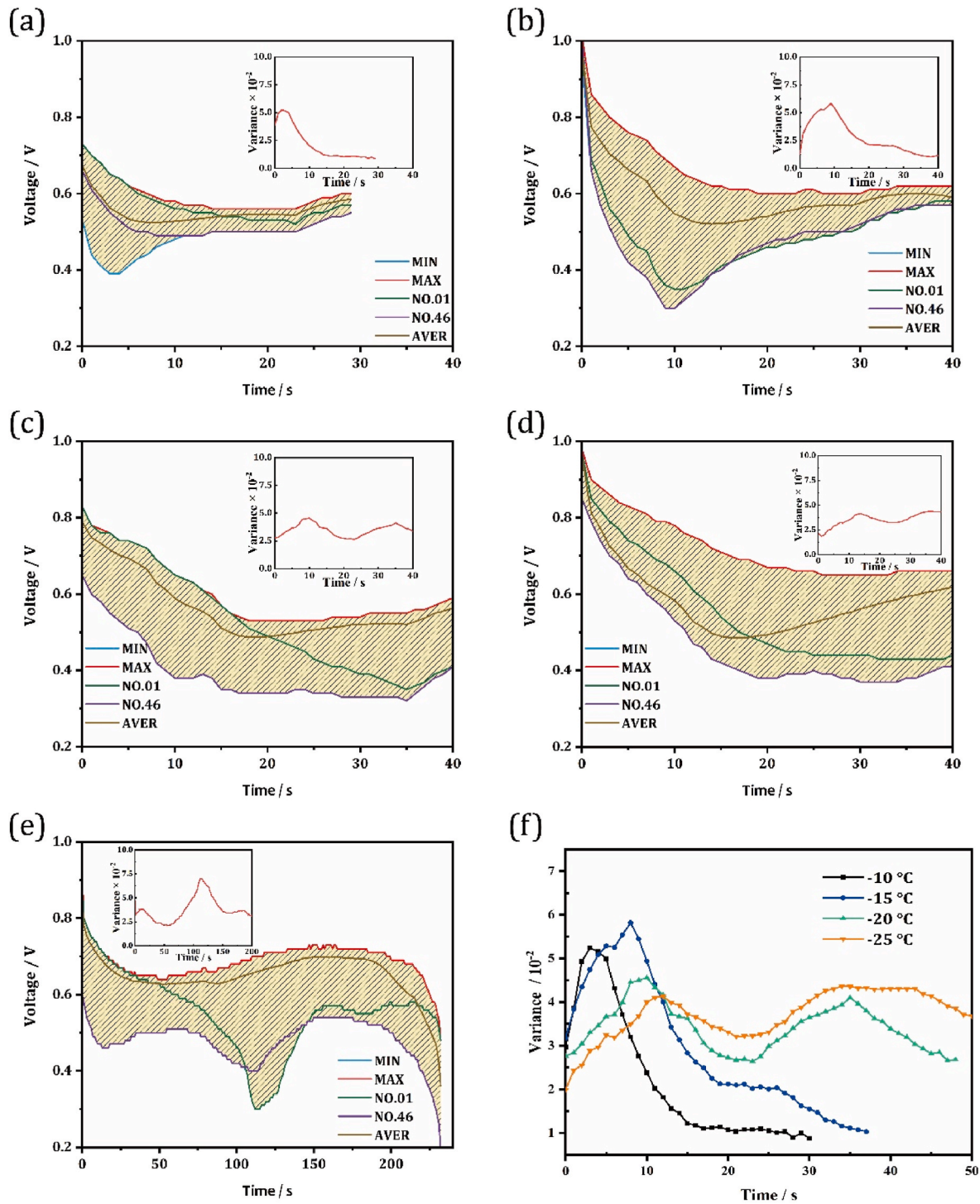


Fig. 4. Voltage consistency of the PEMFC stack during self-cold start from (a) -10°C , (b) -15°C , (c) -20°C , (d) -25°C , (e) -30°C and (f) the voltage variance.

the heating time increases with the decrease of the P_{heater} , which was 51 s for 1.5 kW and 45 s for 3 kW, respectively. As shown in Fig. 6(a), the measured stack temperature continues to grow for a while after the PTC heater was switched off. This is attributed to the residual heat of the heater, and the higher the heating power, the higher the temperature can reach after the heater is turned off. The peak temperature for 3 kW is -15.15°C while is only -17.92°C for 1.5 kW. After reaching the peak temperature, there appear a cooling process due to the temperature decrease after passing through the cold start cooling loop. And the valley temperature decreases with the decrease of the heating power (-15.92°C for 3 kW and -19.93°C for 1.5 kW). Subsequently, the stack

continues to heat up due to the large amount of heat generated by the electrochemical reaction. It worth noting that even through increasing the heating power can shorten the heating time, the improvement is far less than the energy-saving effect brought by lowering the heating power. It's electricity consumption reduced from 0.029 kWh to 0.019 kWh, saving 34.48% of the extra electricity. In addition, the voltage distribution and consistency displayed in Fig. S2 shows that the difference between these two tests are negligible, which further demonstrates that the heating power of 1.5 kW is adequate for cold start of PEMFC stack from -30°C using the proposed strategy.

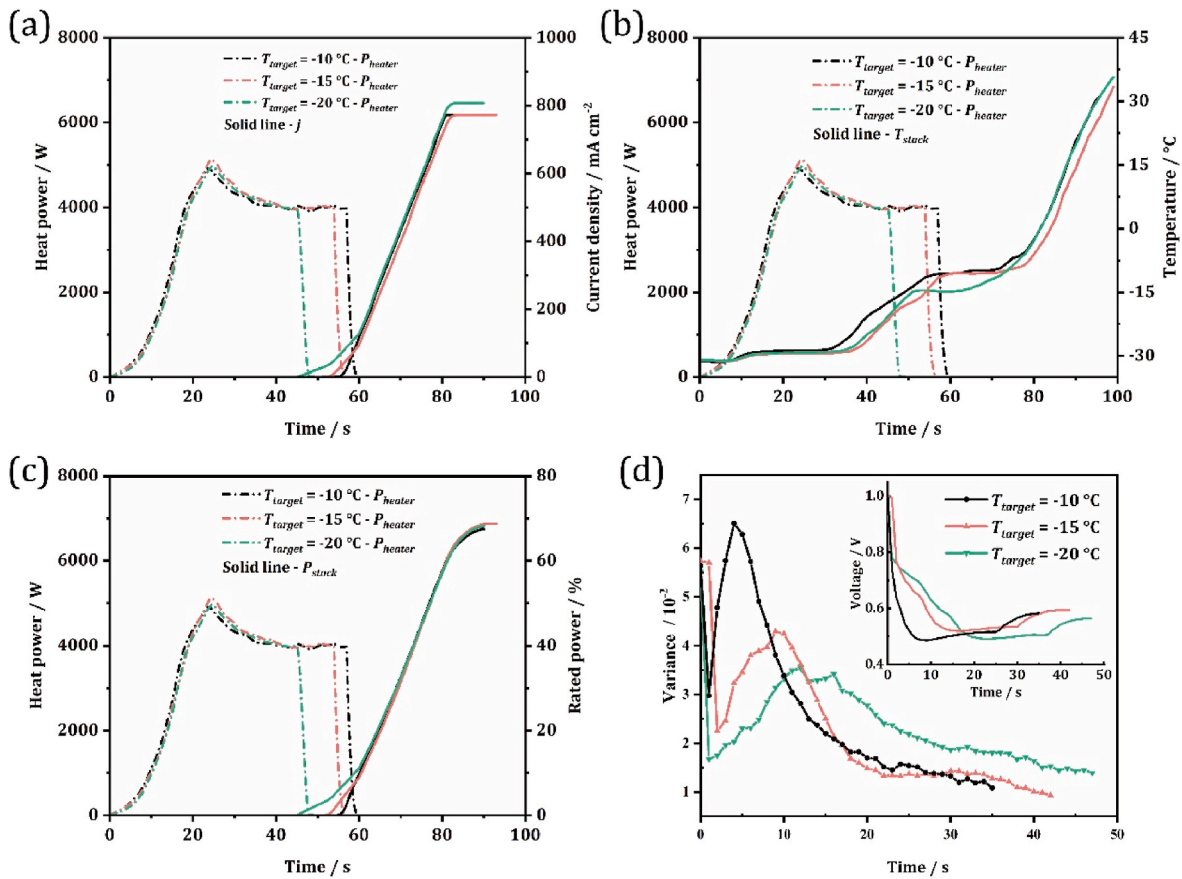


Fig. 5. Effect of target temperature (T_{target}) on the coolant heating assisted cold start of PEMFC stack from -30°C . (a) evolution of the current density; (b) evolution of the stack temperature; (c) evolution of the rated power and (d) evolution of the voltage consistence and the average voltage.

Table 2

Heating time, startup time and electricity consumption of assisted cold start from -30°C .

$T_{stack}/^{\circ}\text{C}$	P_{heater}/kW	$T_{target}/^{\circ}\text{C}$	Heating time/s	Startup time/s	Electricity consumption/kWh
-30	4	-20	45	77	0.039
-30	4	-15	54	78	0.048
-30	4	-10	57	77	0.051
-30	3	-20	45	76	0.029
-30	1.5	-20	51	82	0.019

3.2.2. Cold start from -40°C

Fig. 7 shows the effects of the heating power (P_{heater}) on the cold start performance of the PEMFC stack from -40°C . It can be seen that using this cold start strategy, a successful cold start from -40°C can be achieved within 97 s. The evolution of the cold start process is similar to that of -30°C . However, the difference is that the startup failed when the heating power was 2 kW. As illustrated in Fig. 8(a), the stack reached the peak temperature a few seconds after the heater was switch off, and after that, the stack temperature began to decrease. In order to further explore the characteristics of the cold start with 2-kW heating power, another set of experiment was carried out, during which the PTC heater kept operating until the stack reached the target heating temperature again after passing through the cooling process. The light blue curve and the light green curve in Fig. 8 (a) show that the stack temperature will drop after reaching the peak temperature regardless of whether the heater is operating or not. In this view, it can be concluded that the valley temperature appears because the cooled coolant reenters the stack after passing through the cold-start cooling loop. In other words, during this period, the coolant actually absorbs heat from the originally heated

stack, which is apparently inefficient. Therefore, even though the heating time for the 2-kW was extended to 106 s, the cold start still failed. The voltage distribution and consistency of these tests were shown in Fig. 8. It can be seen that the cold start of the 2-kW case quickly failed after loading and showed poor voltage distribution uniformity and consistency. And the voltage consistency was better with higher heating power as depicted in Fig. 8 (c) and (d). In addition, stack performance tests before and after cold start from -40°C (T_{target} of -20°C and the P_{heater} of 4 kW) shows no signs of degradation as depicted in Fig. S4, which further confirmed the reliability and safety of this strategy. Interestingly, all these cold start tests were started loading until the stack reaches the same measured temperature, why is there such a big difference in the cold start performance?

In order to clarify the relationship between the heating power and cold start performance, we carried out energy balance calculations for the above cold start tests. And the calculations were based on the following assumptions:

- The coolant flow rate is relatively small. Thus, the stack temperature is assumed to be uniform and is measured by the coolant outlet.
- The heat dissipation of the stack and the heat absorbed by the end plate are ignored, because the stack was assembled with insulating plates and the bipolar plates made of metal were relatively thin.
- The latent heat of phase change can be ignored considering that the temperature increased rapidly.
- As the gas went through a cooling pipe before entering the stack and the low heat capacity of the gas, the heat carried by the gas can be ignored.

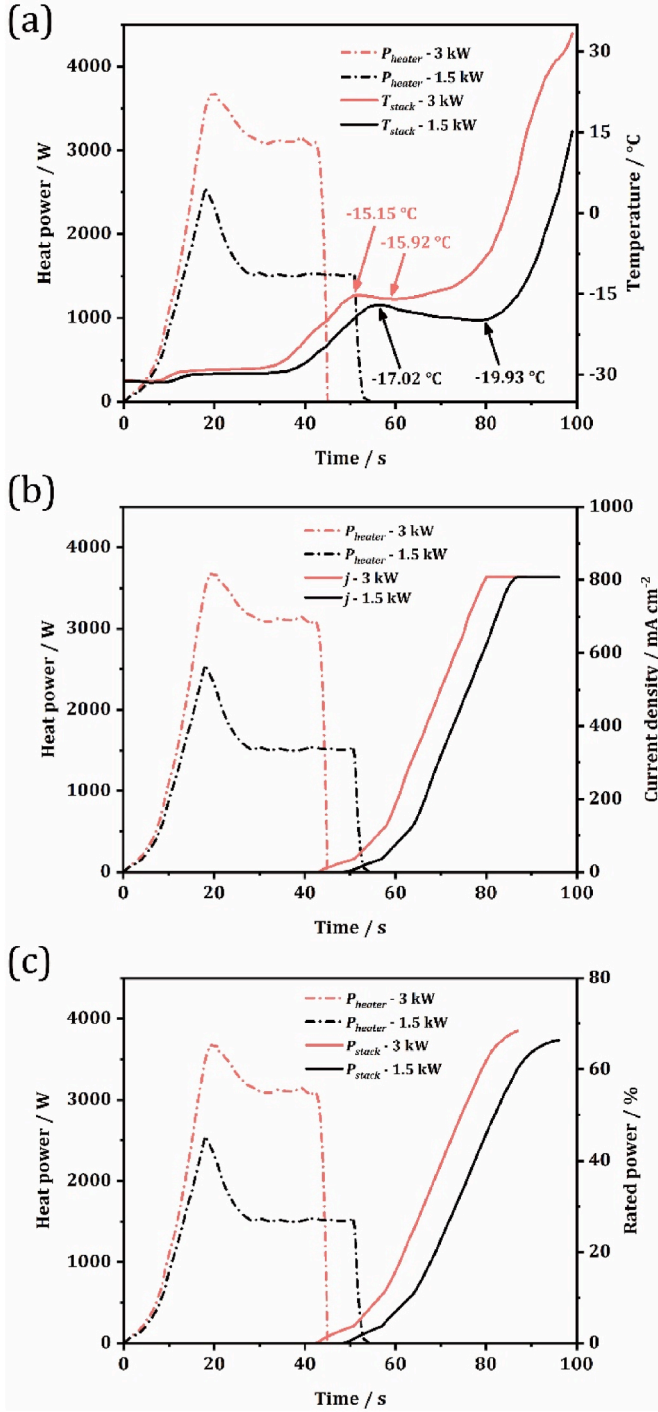


Fig. 6. Effect of heating power (P_{heater}) on the coolant heating assisted cold start of PEMFC stack from $-30 \text{ }^{\circ}\text{C}$. (a) evolution of the stack temperature; (b) evolution of the current density and (c) evolution of the rated power.

(v) The change in density and heat capacity of the coolant with temperature is ignored.

The total heat accumulation rate of the stack consists of two parts. The first is the heat exchange rate between the coolant and the stack. The second is the heat production rate due to the electrochemical reaction.

$$\dot{Q}_{Total} = \dot{Q}_{ex} + \dot{Q}_{ele} \quad (2)$$

The heat exchange rate between the coolant and the stack, $\dot{Q}_{ex}$, can

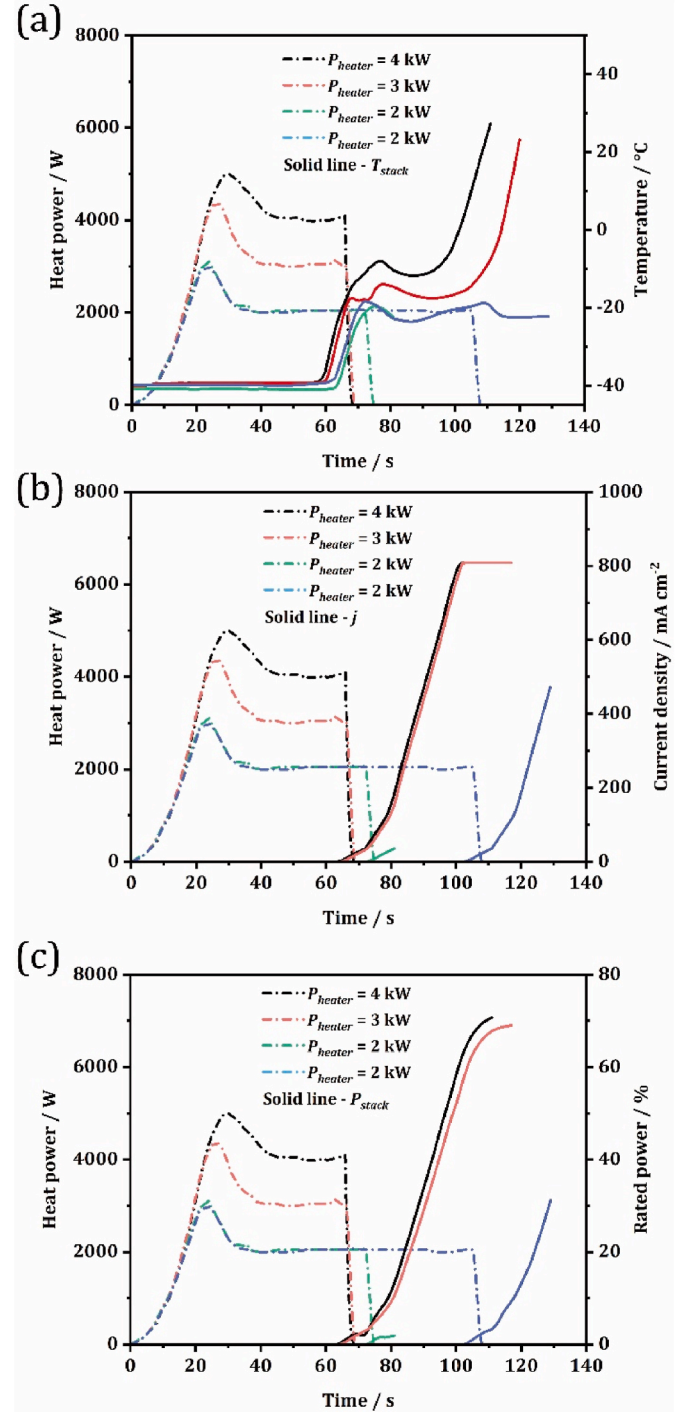


Fig. 7. Effect of heating power (P_{heater}) on the coolant heating assisted cold start of PEMFC stack from $-40 \text{ }^{\circ}\text{C}$. (a) evolution of the stack temperature; (b) evolution of the current density and (c) evolution of the rated power.

be expressed by

$$\dot{Q}_{ex} = (\rho C_p)_{coolant} q_{coolant} (T_{inlet} - T_{outlet}) \quad (3)$$

where $(\rho C_p)_{coolant}$ is the coolant thermal mass, $q_{coolant}$ is the coolant volume flow rate, T_{inlet} is the coolant temperature measured in the inlet of the stack, T_{outlet} is the coolant temperature measured in the outlet of the stack.

The heat production rate due to the electrochemical reaction is calculated from [50].

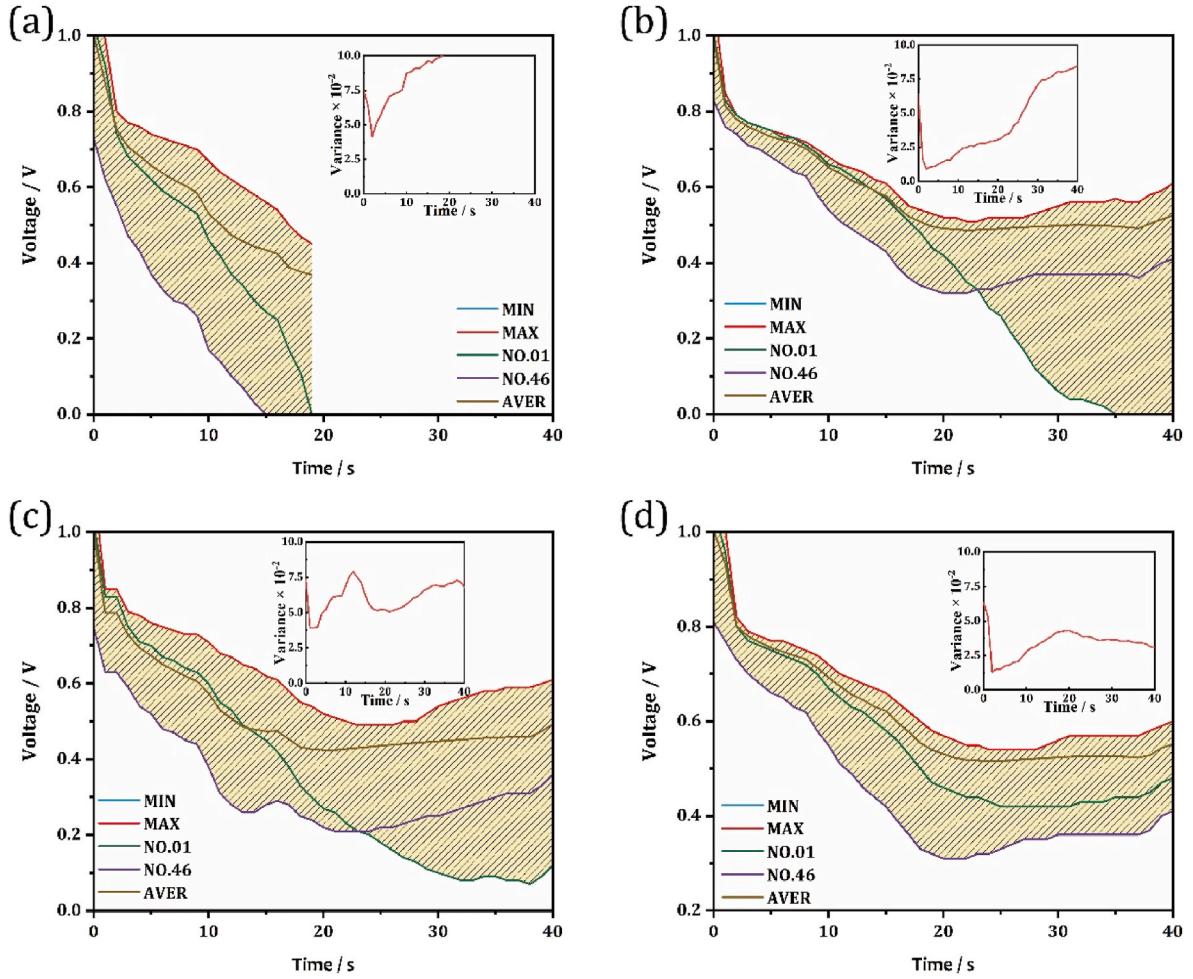


Fig. 8. Voltage distribution and variance during assisted cold start from -40°C with the T_{target} of -20°C and the P_{heater} of (a) 2 kW; (b) 2 kW (with extended heating time); (c) 3 kW and (d) 4 kW.

$$\dot{Q}_{\text{ele}} = (1.48 - U_{\text{cell}}) \cdot I \quad (4)$$

where U_{cell} and I are the measured cell voltage and current. By applying energy balance to the cell and invoking the assumption of a spatially uniform temperature, the cell temperature can be calculated from

$$(mC_P)_{\text{stack}}(T - T_0) = \int_0^t \dot{Q}_{\text{Total}} dt \quad (5)$$

where the stack thermal mass, $(mC_P)_{\text{stack}}$, is about 3.49 kJ/K without including the end plates and insulating plates, but the coolant in the stack is taken into consideration. We calculated the stack temperatures corresponding to three different moments, which correspond to the time when the heater is switched off (T_{switch}), the time when the coolant reaches the peak temperature after the heater is turned off (T_{peak}), and the time when the cold start ends (T_{end}). The calculation results are listed in Table 3.

Table 3
Summary of the heating time, startup time, electricity consumption and calculation results of assisted cold start from -40°C .

$T_{\text{stack}}/$ $^{\circ}\text{C}$	$P_{\text{heater}}/$ kW	$T_{\text{target}}/$ $^{\circ}\text{C}$	Heating time/s	Startup time/s	Electricity consumption/ kWh	$T_{\text{switch}}/^{\circ}\text{C}$		$T_{\text{peak}}/^{\circ}\text{C}$		$T_{\text{end}}/^{\circ}\text{C}$	
						Calculated	Measured	Calculated	Measured	Calculated	Measured
-40	2	-20	73	failed	0.038	-23.44	-20.56	-22.89	-19.66	-21.73	-20.75
-40	2	-20	106	failed	0.056	-21.54	-20.08	-21.35	-18.75	-12.57	-22.31
-40	3	-20	67	100	0.050	-21.05	-20.33	-19.69	-13.93	46.21	43.5
-40	4	-20	66	97	0.060	-18.40	-20.11	-15.00	-7.84	37.43	35.17

As can be seen from Table 3, the calculated temperature by the time the heater was switched off was close to the measured value, indicating high credibility of the measurements and calculations. However, the difference between the calculated value and measured value increases with the extension of time. To explain this, two reasons are considered, the first is that the coolant flow rate was too slow, resulting a certain deviation in the thermal response [33]. The second is that the neglected influence of heat dissipation gradually appears with the extension of time and with the increase of the stack temperature. However, given this stack-level computation, such errors are acceptable. The calculated peak temperature after the heater was switched off were -22.89°C and -21.53°C for the 2-kW cases, respectively, both of which were lower than the target heating temperature. By contrast, the calculated peak temperatures for 3-kW and 4-kW cases were higher than the target heating temperature. And this is responsible for the failed cold start when the P_{heater} was 2 kW while succeed when the P_{heater} were 3 kW and 4 kW. Besides, it can also be explained that although the time to heat the

stack to -20°C under different heating power is close, the actual temperature of the stack is different, which cannot be measured directly by the coolant temperature in the outlet as there is more or less hysteresis in the heat exchange between the coolant and the stack. In addition, the calculated peak temperature for the 4-kW case was 4.69°C higher than the 3-kW case, which also explains the better voltage consistency of cold-starting with the heating power of 4 kW than 3 kW. Moreover, the calculated peak temperature of the 3-kW case is very close to the target heating temperature, and the voltage consistency was relatively poor, which means that 3 kW might be close to the heating power threshold in this experiment. Furthermore, it worth noting that extending the heating time from 76 s to 106 s for the 2-kW case only received limited improvements in cold start performance, while the energy consumption is almost equivalent to that of the 4-kW case. The reason is that during the cooling process after the first peak temperature appears, a large part of the heat is used to heat other components in the cooling loop, such as the coolant pipe, the coolant tank and the coolant recirculation pump, which results in low energy efficiency. Therefore, only when the heating power is matched with the starting temperature and optimized according to the fuel cell system, the fast, safe and efficient characteristics of this cold start strategy can be fully utilized.

4. Conclusions

This paper investigated the cold start of a 7-kW PEMFC stack with metal bipolar plates and proposed an efficient cold start strategy to meet the requirement of the rapid, safe and efficient cold start of PEMFC stack for a wide range of subfreezing temperatures. The proposed cold start strategy combines both the advantages of the self-cold start and coolant heating assisted cold start. To this end, we first explored the self-cold start ability of the metal bipolar plate PEMFC stack. Then, the characteristics the cold start strategy was investigated in terms of the effects of the stack temperature, the heating power, the target heating temperature. In addition, key parameters during cold start such as the startup time, the energy consumption, safety and the voltage consistency are analyzed in detail.

1. PEMFC stack with metal bipolar plates exhibits excellent self-cold start performance. A rapid and safe self-start from -20°C and -25°C can be achieved within 30 s and 57 s, respectively. Besides, voltage consistency analyze indicates that there exists serious end plate effect during cold start and the voltage consistency during self-cold start deteriorates with the decrease of startup temperature.
2. Considering the startup time, safety, energy consumption and voltage consistency, target heating temperature of -20°C was adequate for the proposed cold start strategy.
3. The proposed start-up strategy is capable of achieving rapid and safe cold start from -30°C and -40°C within 82 s and 100 s with the electricity consumption is 0.019 kWh and 0.050 kWh, respectively.
4. Energy balance calculation indicates that only when the stack peak temperature after the heater is switch off is higher than the target heating temperature, the cold start can be successful. Besides, the higher the heating power, the better the voltage consistency during cold start. In addition, it is necessary to avoid using excessive heating power as the improvement of cold start performance may be insignificant, but the increased energy consumption is significant. At the same time, the use of insufficient heating power must also be avoided, as it may cause start-up failure, and even prolonged heating time can be inefficient.

CRediT authorship contribution statement

Xiaokang Yang: Conceptualization, Methodology, Investigation, Writing – original draft. **Jiaqi Sun:** Software, Writing – review & editing. **Shucheng Sun:** Writing – original draft, Supervision. **Zhigang Shao:** Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support from “Transformational Technologies for Clean Energy and Demonstration”, Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDA21090101), and the Liaoning Revitalization Talents Program (Grant No. XLYC1902079). The authors would also like to thank the reviewers for their valuable comments and suggestions.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2022.231492>.

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